

EXP 2: COMPUTATION OF TABLE AND GRAPHS

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```
empid=c(1,2,3,4,5,6,7,8,9,10)
empid #for printing
age=c(30,25,40,34,55,50,35,23,54,44)
age
gender=c(1,0,1,0,1,1,1,1,0,1)
gender
status=c(1,1,2,2,1,2,1,2,2,1)
status
```

```
empinfo=data.frame(empid,age,gender,status)
empinfo
```

```
nrow(empinfo)
ncol(empinfo)
```

```
empinfo$gender=factor(empinfo$gender,levels=c(0,1),labels=c("Male","Female"))
empinfo
empinfo$status=factor(empinfo$status,levels=c(1,2),labels=c("Staff","Faculty"))
empinfo
summary(empinfo)
```

```
> Sys.time()
[1] "2026-07-14 09:03:25 IST"
> empid=c(1,2,3,4,5,6,7,8,9,10)
> empid #for printing
[1] 1 2 3 4 5 6 7 8 9 10
> age=c(30,25,40,34,55,50,35,23,54,44)
> age
[1] 30 25 40 34 55 50 35 23 54 44
> gender=c(1,0,1,0,1,1,1,1,0,1)
> gender
[1] 1 0 1 0 1 1 1 1 0 1
> status=c(1,1,2,2,1,2,1,2,2,1)
> status
[1] 1 1 2 2 1 2 1 2 2 1
> empinfo=data.frame(empid,age,gender,status)
> empinfo
  empid age gender status
1     1  30     1      1
2     2  25     0      1
3     3  40     1      2
4     4  34     0      2
5     5  55     1      1
6     6  50     1      2
7     7  35     1      1
8     8  23     1      2
9     9  54     0      2
10    10  44     1      1
```

```
> nrow(empinfo)
[1] 10
> ncol(empinfo)
[1] 4
```

```
> empinfo
  empid age gender status
1     1  30 Female  Staff
2     2  25  Male  Staff
3     3  40 Female Faculty
4     4  34  Male Faculty
5     5  55 Female  Staff
6     6  50 Female Faculty
7     7  35 Female  Staff
8     8  23 Female Faculty
9     9  54  Male Faculty
10    10  44 Female  Staff
```

```
> summary(empinfo)
      empid          age          gender          status
Min.   : 1.00   Min.   :23.0   Male   :3   Staff   :5
1st Qu.: 3.25   1st Qu.:31.0   Female:7   Faculty:5
Median : 5.50   Median :37.5
Mean   : 5.50   Mean   :39.0
3rd Qu.: 7.75   3rd Qu.:48.5
Max.   :10.00   Max.   :55.0
```

```
empinfo$age #display only the age column
```

```
empinfo[, "age", drop=FALSE] #display as column vector
```

```
empinfo[,1]
```

```
empinfo[,2]
```

```
empinfo[1,]
```

```
empinfo[2,]
```

```
empinfo[3,3]
```

```
empinfo[1:5,2:3]
```

```

> empinfo$age #display only the age column
[1] 30 25 40 34 55 50 35 23 54 44
> empinfo[,"age",drop=FALSE] #display as column vector
  age
1   30
2   25
3   40
4   34
5   55
6   50
7   35
8   23
9   54
10  44
> empinfo[,1]
[1] 1 2 3 4 5 6 7 8 9 10
> empinfo[,2]
[1] 30 25 40 34 55 50 35 23 54 44
> empinfo[1,]
  empid age gender status
1     1  30 Female  Staff
> empinfo[2,]
  empid age gender status
2     2  25  Male  Staff

```

```

> empinfo[3,3]
[1] Female
Levels: Male Female

```

```

> empinfo[1:5,2:3]
  age gender
1  30 Female
2  25  Male
3  40 Female
4  34  Male
5  55 Female

```

```

plot(age,type="o",main="Age of subjects",
     xlab="Employee order",
     ylab="Age in years",
     col="blue")

```

```

#pie plot
Table1=table(empinfo$gender);
Table1
pie(Table1)

```

```

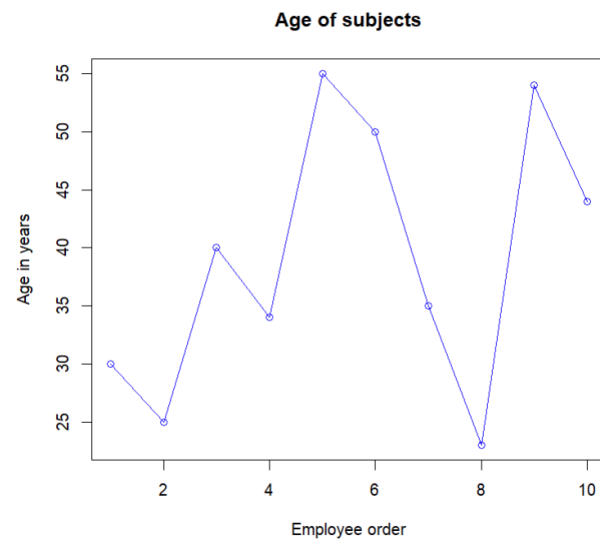
#Bar plot
Table3=table(empinfo$gender,empinfo$status)

```

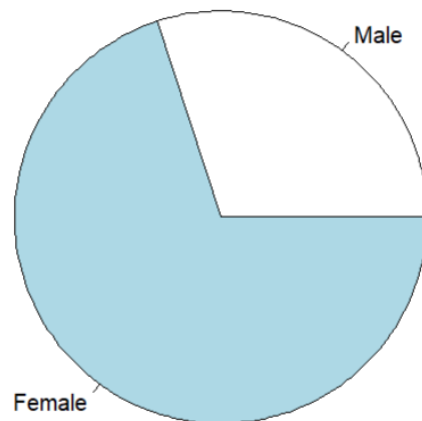
Table3

```
barplot(Table3,beside=T,xlim=c(1,15),ylim=c(0,5),col=c("blue","red"))
legend("topright",legend=rownames(Table3),fill=c("blue","red"))
```

```
col=„purple„)
λjap=„Age in years„,
xjap=„Employee order„,
bjot(αδ6, fλb6=„o„, wαt=„Age of subjects„,
```



```
pie(Table1)
```



```
barplot(Table3,beside=T,xlim=c(1,15),ylim=c(0,5),col=c("blue","red"))
legend("topright",legend=rownames(Table3),fill=c("blue","red"))
```

